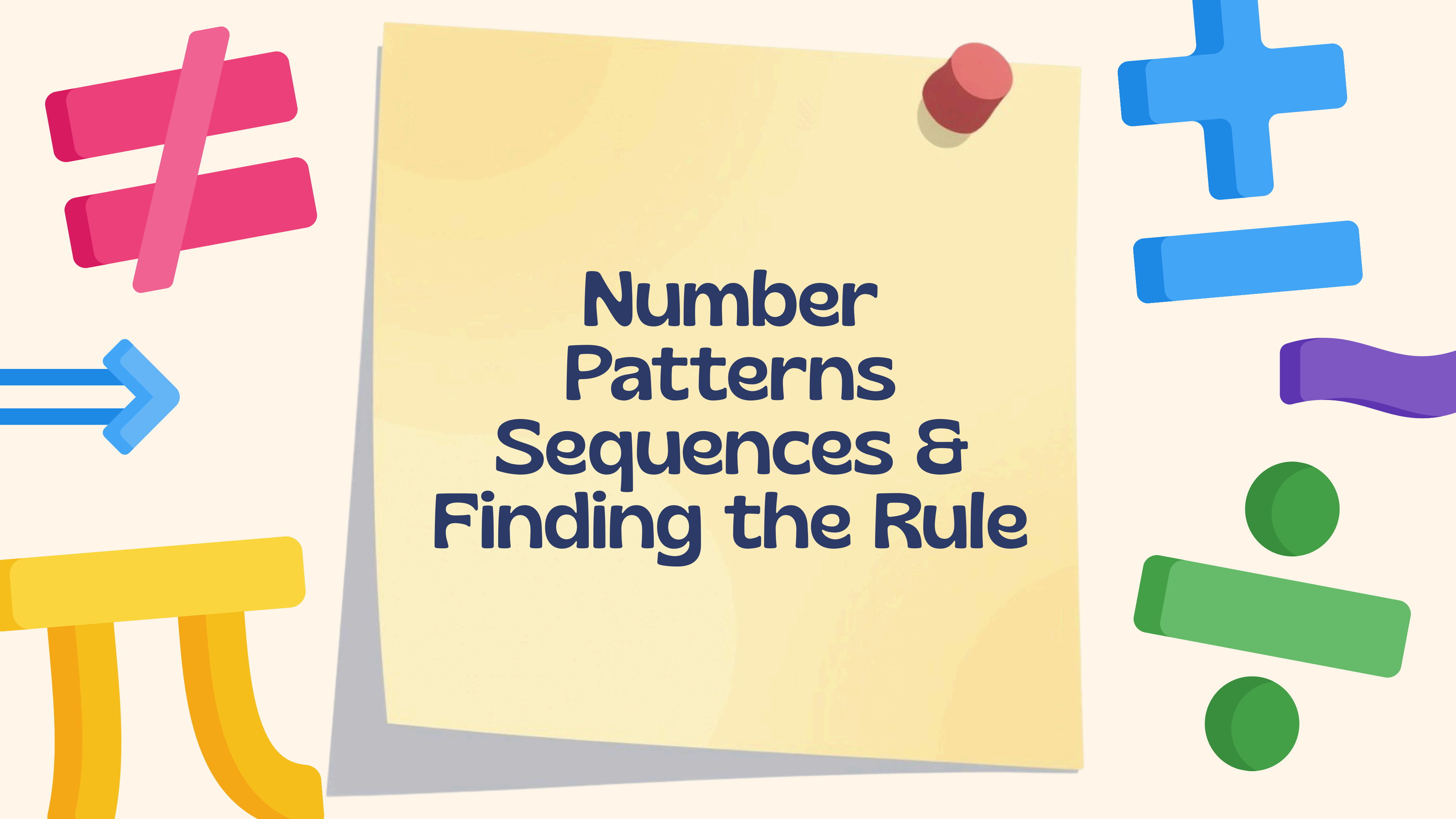
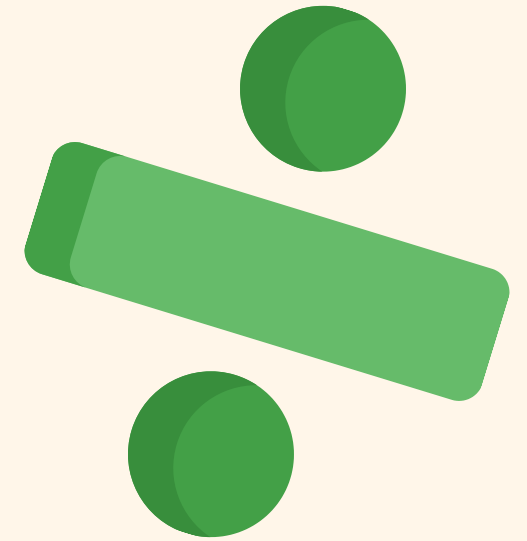
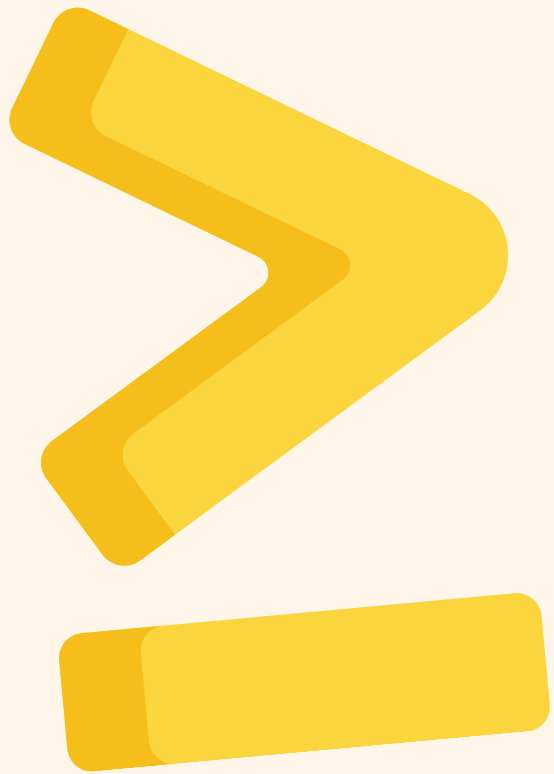




Number Patterns Sequences & Finding the Rule

What Are Number Patterns?

- A number pattern is a list of numbers that follow a special rule. Think of it like a recipe where each step tells you how to get the next number!
- A sequence is just numbers in a special order. For example, look at this sequence: 2, 4, 6, 8...
- Notice what happens? Each number goes up by 2! That's the rule. The arrows help us see how the numbers grow from one to the next.
- Patterns are everywhere - they help us predict what comes next and make learning math fun!



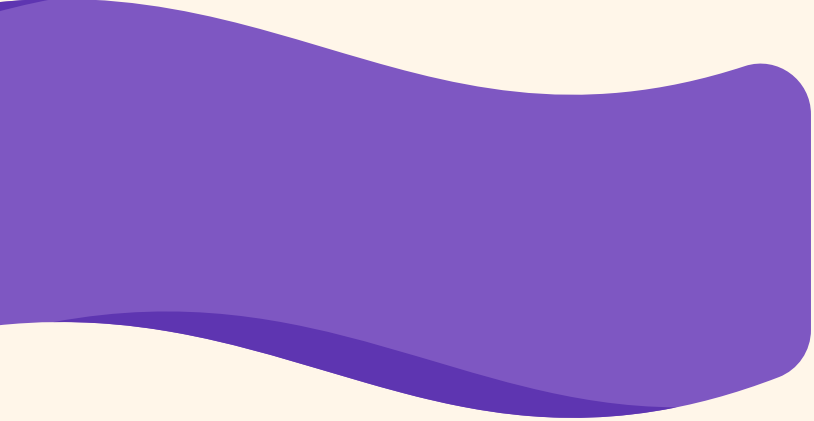


Examples of Number Sequences



- 1, 3, 5, 7... → Add 2 each time! Each number increases by 2.
- 10, 9, 8, 7... → Subtract 1 each time! Each number goes down by 1.
- 5, 10, 15, 20... → Add 5 each time! Each number increases by 5.

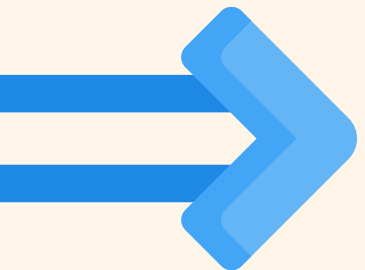




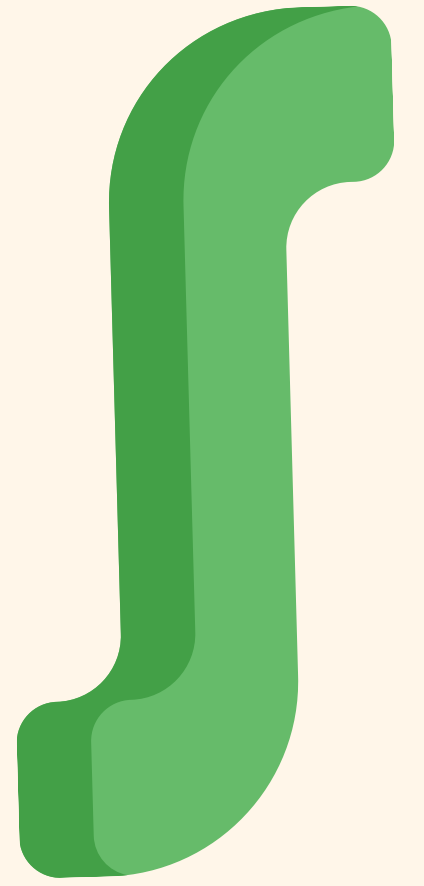
How to Find the Rule



1. Look carefully at the numbers in the sequence.
 2. Find how much the number changes each time - is it adding, subtracting, or something else?
 3. Say the rule in words!
- 
- 
- 



Example: Look at 3, 6, 9, 12... Each number goes up by 3! So the rule is "Add 3 each time."
Practice this with your own number patterns!





Growing & Shrinking Patterns

Growing

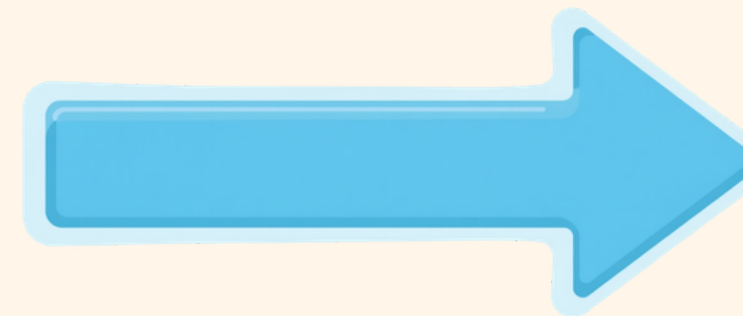
Growing patterns get bigger step by step! Each number is larger than the one before it. Example: 2, 4, 8, 16 - we double each time! ($\times 2$)

Middle

The middle shows the pattern rule. Can you spot the rule? Growing patterns add or multiply to get bigger. Shrinking patterns subtract or divide to get smaller.

Shrinking

Shrinking patterns get smaller step by step! Each number is smaller than the one before it. Example: 100, 90, 80, 70 - we subtract 10 each time! (-10)





The position number tells you how to find the number! Each number in a sequence has a special position - 1st, 2nd, 3rd, and so on. Let's look at a fun example together!

5

Look at this sequence: 5, 10, 15, 20. The position rule is: $\text{nth term} = 5 \times n$. This means the first number is $5 \times 1 = 5$, the second is $5 \times 2 = 10$, the third is $5 \times 3 = 15$, and so on!

Position Rules



Let's use arrows to show how positions work: 1st $\rightarrow$ 5, 2nd $\rightarrow$ 10, 3rd $\rightarrow$ 15, 4th $\rightarrow$ 20. To find any number, just multiply the position number by 5! This is called the 'nth term rule' and it helps us find any number in the sequence without counting!





Be a Pattern Detective!



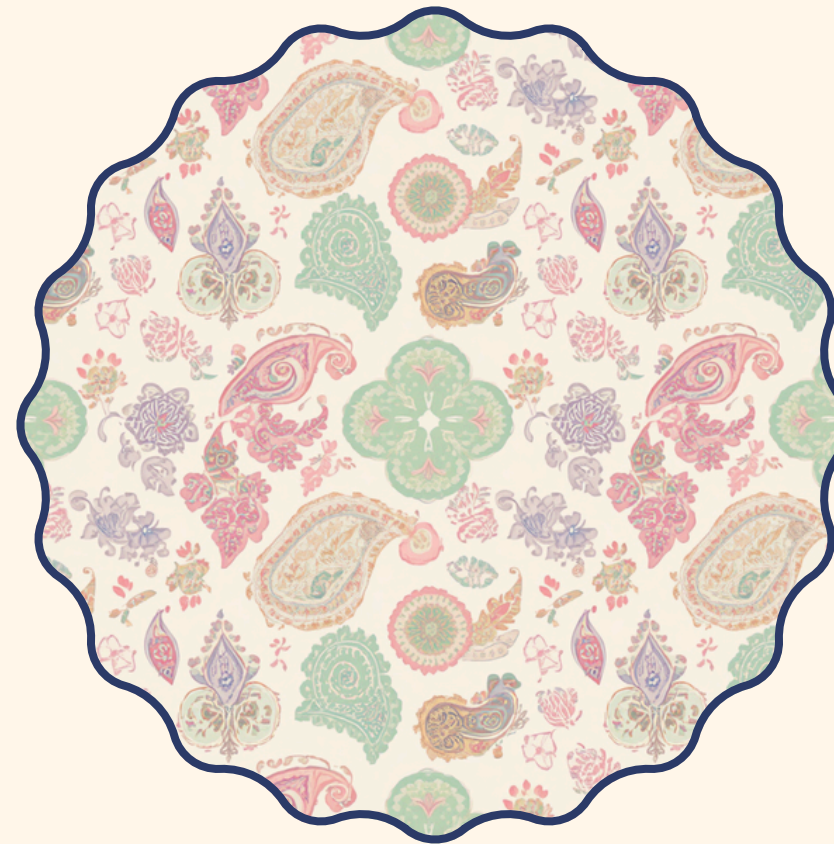
Look at this number sequence: 3, 6, __, 12, __, 18
Can you find the missing numbers? What's the rule?
Think like a detective and solve the mystery!

Hint: Each number changes by the same amount. Once
you find the rule, you can fill in the blanks!





Real-World Indian Number Patterns





Practice Exercises



2, __, 6, __, 10

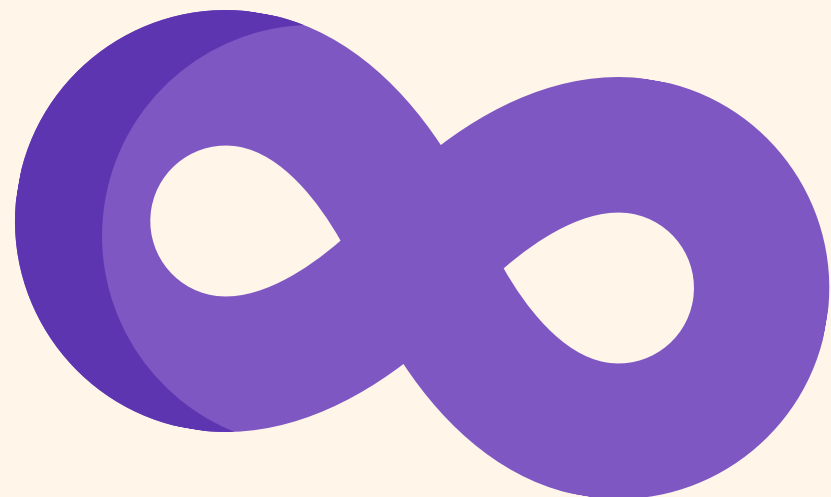
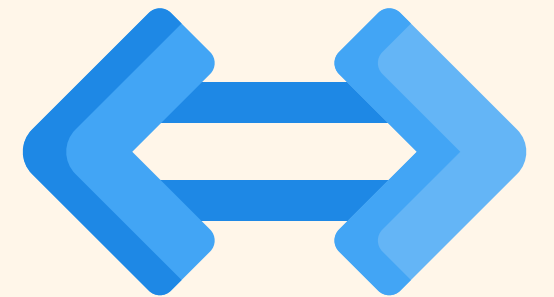
Rule: Add 2 each time

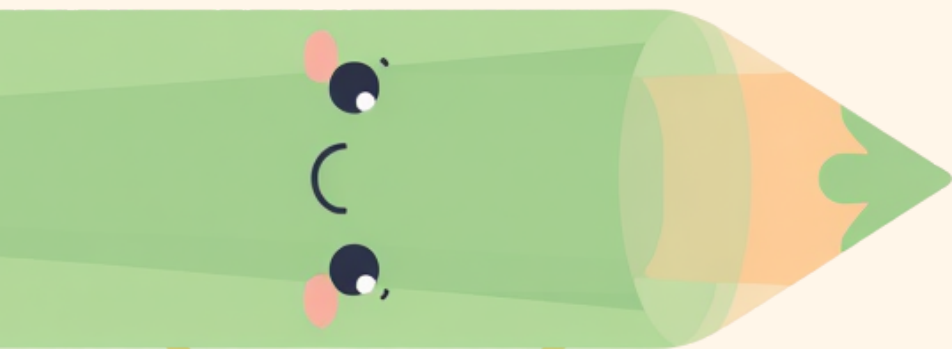
20, 18, __, 14, __

Rule: Subtract 2 each time

4, 8, 12, __, __

Rule: Add 4 each time





**Well Done!
Keep Exploring
Patterns!**

